

Total No. of Questions : 4]

SEAT No. :

P8473

[Total No. of Pages : 1

Oct-22/BE/Insem-52

B.E. (Computer Engineering)

SOFTWARE TESTING AND QUALITY ASSURANCE
(2019 Pattern) (Semester - VII) (Elective-IV) (410245-D)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data if necessary.

- Q1)** a) Define software quality. List & explain core component of quality. [5]
b) What are the constraints of software product quality assesement. [5]
c) Explain defect life cycle with the help of diagram. [5]

OR

- Q2)** a) Examine the relationship between quality & productivity. [5]
b) Explain PDCA cycle. [5]
c) Plan software quality control with respect to college attendance software. [5]

- Q3)** a) Write test cases for login validation. [5]
b) What are the entry & exit criteria of testing. [5]
c) Analyse test policy & test strategy which is included in test documentation. [5]

OR

- Q4)** a) Explain use case testing with one example. [5]
b) How can you classify the role of software [5]
c) Differentiate between test plan & test strategy. [5]



Total No. of Questions : 4]

SEAT No. :

P8473

[Total No. of Pages : 1

Oct-22/BE/Insem-52

B.E. (Computer Engineering)

**SOFTWARE TESTING AND QUALITY ASSURANCE
(2019 Pattern) (Semester - VII) (Elective-IV) (410245-D)**

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data if necessary.

- Q1)** a) Define software quality. List & explain core component of quality. [5]
b) What are the constraints of software product quality assesement. [5]
c) Explain defect life cycle with the help of diagram. [5]

OR

- Q2)** a) Examine the relationship between quality & productivity. [5]
b) Explain PDCA cycle. [5]
c) Plan software quality control with respect to college attendance software. [5]

- Q3)** a) Write test cases for login validation. [5]
b) What are the entry & exit criteria of testing. [5]
c) Analyse test policy & test strategy which is included in test documentation. [5]

OR

- Q4)** a) Explain use case testing with one example. [5]
b) How can you classify the role of software [5]
c) Differentiate between test plan & test strategy. [5]



Total No. of Questions : 4]

SEAT No. :

PC-194

[Total No. of Page : 1

[6361]-52

B.E. (Computer) (Insem.)

**SOFTWARE TESTING & QUALITY ASSURANCE
(2019 Pattern) (Semester - VII) (410245 - D) (Elective - IV)**

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates :

- 1) *Solve Q.1 or Q.2, Q.3 or Q.4.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Assume suitable data, if necessary.*
- 4) *Figures to the right indicate full marks.*

- Q1)** a) Construct SDLC for any real life application. [5]
b) Write a note on pillars of Quality Management System. [5]
c) What is the difference between Testing & Debugging. [5]

OR

- Q2)** a) Explain the problematic area of software development life cycle. [5]
b) List and Explain Quality objectives applicable to software development and usage. [5]
c) Plan Software Quality Control with respect to college attendance software. [5]
- Q3)** a) Explain the test efficiency and defect rejection. [5]
b) What are the entry & exit criteria of testing. [5]
c) Describe cultural changes requirement for Quality improvement. [5]

OR

- Q4)** a) Explain use case testing with one example. [5]
b) Differentiate between Test plan and Test strategy. [5]
c) Describe Quality Assurance Process. [5]



Total No. of Questions : 4]

SEAT No. :

P8466

[Total No. of Pages : 2

Oct-22/BE/Insem-42

B.E. (Computer Engg.)

MACHINE LEARNING

(2019 Pattern) (Semester - VII) (410242)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates :

- 1) Answer Q1 or Q2, Q3 or Q4.
- 2) Neat diagrams must be drawn whenever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

Q1) a) Compare machine learning vs Artificial Intelligence. [5]

b) Describe parametric and Non-parametric machine learning models. [5]

c) Explain various Data formats that conform ML elements. [5]

OR

Q2) a) Explain supervised, unsupervised and semi supervised learning. [7]

b) Describe various statistical learning approaches. [8]

Q3) a) Consider a vector $x = (23, 29, 52, 31, 45, 19, 18, 27)$ Apply feature scaling and find out min-max scaled values as well as z-score values. [8]

b) Explain the process of Principal Component Analysis (PCA) in brief. [7]

OR

P.T.O.

- Q4)** a) How to handle missing values in a dataset that will be used for training the ML model? [5]
- b) Explain the types of wrapper methods for feature selection. [5]
- c) Explain Local Binary Pattern (LBP) feature extraction technique with suitable example. [5]

Total No. of Questions : 4]

SEAT No. :

P5190

[Total No. of Pages : 2

[6188]-142

**B.E. (Computer Engineering) (Insem)
MACHINE LEARNING
(2019 Pattern) (Semester - VII) (410242)**

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4.
- 2) Figures to the right side indicate full marks.
- 3) Draw neat diagrams wherever necessary.
- 4) Assume suitable data if necessary.

- Q1)** a) Compare Machine Learning with traditional programming. Discuss types of Machine Learning with suitable examples. [5]
- b) What are various Statistical Learning Approaches? [5]
- c) Explain different dataformats used in Machine Learning. [5]

OR

- Q2)** a) What is Machine Learning? Explain applications of Machine Learning in data science. [5]
- b) Explain Geometric Model and Probabilistic Model with suitable examples. [5]
- c) How machine learning model works? Explain various steps involved. [5]

- Q3)** a) What is feature selection? Explain filtering technique [5]
- b) Explain kernel PCA in detail. [5]
- c) Calculate LBP code generated value for the central point in the neighborhood of 8 pixels as shown below. [5]

10	12	18
7	9	6
9	2	4

OR

P.T.O.

- Q4)** a) Explain Min-Max scaling with suitable example. [5]
- b) What is Matrix factorization? Explain content based filtering with an example. [5]
- c) Given following data for attribute AGE calculate Z- score normalization. [5]
- AGE = {18, 22, 25, 42, 28, 43, 33, 35, 56, 28}



Total No. of Questions : 4]

SEAT No. :

PC184

[6361]-42

[Total No. of Pages : 1

B.E. (Computer Engineering) (Insem)
MACHINE LEARNING
(2019 Pattern) (Semester - VII) (410242)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4.
- 2) Figures to the right indicates full marks.
- 3) Draw neat diagram wherever necessary.
- 4) Assume suitable data, if necessary.

- Q1)** a) Describe in detail different Machine Learning models used. [5]
b) What are the main differences between supervised learning and reinforcement learning? [5]
c) Explain geometric models and its types. [5]

OR

- Q2)** a) Compare Artificial intelligence and Machine learning. [5]
b) Describe the various steps involved in developing a machine learning application. [5]
c) Describe grouping and grading models. [5]
- Q3)** a) Convert given data set in normalized data set. $D = \{23, 29, 52, 31, 45, 19, 18, 27\}$. [8]
b) Why do you need categorical variable encoding? With an example, explain one-hot encoding. [7]

OR

- Q4)** a) Which statistical methods are used to describe the nature of data? [5]
b) What are the feature of multidimensional scaling? [5]
c) Elaborate use of PCA in preprocessing stage. [5]

* * *

Total No. of Questions : 4]

SEAT No. :

P8467

[Total No. of Pages : 1

Oct-22/BE/Insem-43

**B.E. (Computer Engineering)
BLOCKCHAIN TECHNOLOGY
(2019 Pattern) (Semester-VII) (410243)**

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

- Q1)** a) List and explain advantages of ECC. [5]
b) Differentiate asymmetric and symmetric key cryptography. [5]
c) Discuss the properties of hash function. [5]

OR

- Q2)** a) List and discuss benefits of Merkel tree. [5]
b) Explain DSA key generation and verification. [5]
c) Discuss role of hashing in Blockchain. [5]

- Q3)** a) Explain consensus layer in Blockchain. [5]
b) Discuss limitations of centralised systems. [5]
c) Why Blockchain is important? [5]

OR

- Q4)** a) Explain propagation layer in blockchain. [5]
b) Discuss evolution of Blockchain. [5]
c) Differentiate centralised & decentralised systems. [5]



Total No. of Questions : 4]

SEAT No. :

P-5191

[Total No. of Pages : 1

[6188]-143

B.E. (Computer Engineering) (Insem)
BLOCK CHAIN TECHNOLOGY
(2019 Pattern) (Semester - VII) (410243)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) *Solve Q1 or Q2, Q3 or Q4.*
- 2) *Figures to the right indicate full marks.*
- 3) *Assume Suitable data, if necessary.*
- 4) *Neat diagrams must be drawn wherever necessary.*

- Q1)** a) Illustrate Elliptic Curve Cryptography. [6]
b) Justify the importance of Hashing in Block Chain. [4]
c) What is Merkle tree? Explain the structure of merkle tree. [5]

OR

- Q2)** a) Explain working of SHA 256 Algorithm. [6]
b) Describe Digital Signature & Verification steps in Digital Signature Algorithm. [4]
c) Describe Symmetric Key Encryption with neat diagram. [5]

- Q3)** a) Discuss various limitations of centralized system with respect to Decentralized system. [6]
b) Write a note on Evolution of Block Chain. [4]
c) List and Explain algorithms of Consensus layer. [5]

OR

- Q4)** a) Write a note on : [6]
i) Propagation layer
ii) Application layer
b) List & Explain features of Block Chain. [4]
c) Comment on "Feasibility of an Online Voting System Implementation" using Block Chain Technology. [5]



Total No. of Questions : 4]

SEAT No. :

PC185

[6361]-43

[Total No. of Pages : 1

B.E. (Computer Engineering) (Insem)
BLOCKCHAIN TECHNOLOGY
(2019 Pattern) (Semester-VII) (410243)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) *Answer Q.1 or Q.2, Q.3 or Q.4.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Assume suitable data if necessary.*

- Q1)** a) Differentiate between symmetric and asymmetric key cryptography. [5]
- b) What is hashing? Explain role of hashing in Blockchain. [5]
- c) What is Merkle tree? Explain with diagram. [5]

OR

- Q2)** a) Explain digital signature algorithm. [5]
- b) What are the benefits of Merkle tree in Blockchain. [5]
- c) Discuss elliptic curve cryptography. [5]

- Q3)** a) Explain consensus Layer in Blockchain. [5]
- b) Differentiate between centralized and decentralized systems. [5]
- c) Discuss evolution of Blockchain. [5]

OR

- Q4)** a) Explain semantic Layer in Blockchain. [5]
- b) Discuss importance of Blockchain. [5]
- c) What is decentralized system? Explain with diagram. [5]



Total No. of Questions : 4]

SEAT No. :

P8465

[Total No. of Pages : 2

Oct-22/BE/Insem-41

B.E. (Computer Engineering)

DESIGN AND ANALYSIS OF ALGORITHMS

(2019 Pattern) (Semester - VII) (410241)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer the question of Q.1 or Q.2, Q.3 or Q.4.
- 2) Neat diagrams must be drawn whenever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

Q1) a) Why correctness of the algorithm is important? Define loop invariant property and prove the correctness of finding summation of n numbers using loop invariant property. [8]

b) What is iterative algorithm? Explain iterative algorithm design issues using examples. [7]

OR

Q2) a) How to prove that an algorithm is correct? How to prove the correctness of an algorithm using counter example? Give suitable example. [7]

b) Write a short note on any 4 problem solving strategies. [8]

Q3) a) What is Best, Average and Worst case Analysis of Algorithms? Analyse the following algorithm Best, Average and Worst case [8]

```
void sort (int a, int n) {
    int i, j;
    for (i = 0; i < n; i++) {
        j = i-1;
        key = a[i];
        while (j >= 0 && a[j] > key)
        {
            a[j+1] = a[j];
            j = j-1;
        }
        a[j+1] = key;
    }
}
```

P.T.O.

- b) • Explain P, NP, NP-Hard and NP-Complete problems with examples.
- Explain 3-SAT problem using an example. Why is SAT so important in theoretical computer science?

[7]

OR

Q4) a) What is NP-complete class problem? How would you prove vertex cover problem is NP-complete class problem? [8]

b) What is Best, Average and Worst case Analysis of Algorithms? Analyse the following algorithm Best, Average and Worst case [7]

```
int Linear-search(int a, int n, int item) {  
    int i;  
    for (i = 0; i < n; i++) {  
        if (a[i] == item) {  
            return a[i]  
        }  
    }  
    return - 1  
}
```

Total No. of Questions : 4]

SEAT No. :

P5189

[Total No. of Pages : 2

[6188]:141

B.E. (Computer Engineering) (Insem)
DESIGN AND ANALYSIS OF ALGORITHMS
(2019 Pattern) (Semester-VII) (410241)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer the questions Q.1 or Q.2; Q.3 or Q.4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

- Q1)** a) Given the fastest computer and hypothetically infinite memory, do we still need to study algorithms? Justify. [2]
- b) How can we related algorithms to technology? Briefly explain. [6]
- c) Consider an array A of n integers which are already in sorted order. Let x be the number being searched in the array A in a liner fashion. The code fragment performing this task is given below: [7]

```
int lin _ search (int A [])
{
    i=0; flag=0;
    do { if (x == A [i]) then
        return (1); // Number found
        else
            i++;
    } while (i<n);
    return (0); // Number not found.
}
```

- i) Is this code fragment efficient? (We wish to use linear search only). Justify your answer.
- ii) Does it attribute to any design issue with respect to iterative algorithm? Briefly explain.

OR

P.T.O.

Q2) a) What is iterative algorithm? Explain iterative algorithm design issues using suitable examples. [8]

b) Consider the following algorithm to find the square of a number:

```
int sqr(int n)
{
    if (n == 0) return 0;
    else return (2n+sqr(n-1)-1)
```

Prove the correctness of this algorithm using principle of mathematical induction or otherwise. [7]

Q3) a) Briefly explain P and NP problems in the context of complexity theory. Give suitable example. [8]

b) If $f(n) = O(g(n))$ then does it imply $g(n) = O(f(n))$? Discuss. [5]

c) Comment on the statement “Best case analysis of algorithm may not give clear idea of performance” [2]

OR

Q4) a) What is SAT AND 3-SAT problem? Prove that 3-SAT problem is NP complete. [8]

b) What do you understand by best case, worst case and average-case behaviour of an algorithm? Is an average case efficiency an average of best-case, worst-case efficiencies? Justify answer. [7]



Total No. of Questions : 4]

SEAT No. :

PC183

[6361] 41

[Total No. of Pages : 2

B.E. (Computer Engineering) (Insem)
DESIGN AND ANALYSIS OF ALGORITHMS
(2019 Pattern) (Semester - VII) (410241)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figure to right indicate full marks.
- 4) Assume suitable data, if necessary.

- Q1) a)** Prove the correctness of the following algorithm, which finds the minimum number from the given array of n numbers.

```
int min_element (int A [ ])
```

```
{
```

```
    min=A[0];
```

```
    for (i=1; i<n ; i++)
```

```
    {
```

```
        if (A[i]<min) then
```

```
            min=A[i];
```

```
    }
```

```
    return(min);
```

```
}
```

[7]

- b) What is iterative algorithm? Explain iterative algorithm design issues using examples. [8]

OR

- Q2) a)** Prove the correctness of the following algorithm, which finds the factorial of a given number.

```
int fact (int n)
```

```
{ if (n==0) then return 1;
```

```
  else
```

```
    return (n * fact(n-1));
```

```
}
```

[7]

- b) Write a short note on any four problem solving strategies. [8]

P.T.O.

Q3) a) Consider the algorithm for selection sort:

```
void select (int A[ ])
{
    // A is an array of n numbers.
    for (i=1; i<n-1; i++)
    {
        min_pos=i;
        min_val=A[i];
        for(j=i+1; j<=n; j++)
        {
            if(A[j] < min_val) then
            {
                min_pos=j;
                min_val=A[j];
            }
        }
        A[min_pos]=A[i];
        A[i]=min_val;
    }
}
```

Analyse the complexity of this algorithm. Clearly indicate the assumptions made, if any. **[6]**

- b) If an algorithm has a time complexity of both $O(f(n))$ and $\Omega(g(n))$ for the same function $f(n)$ and $g(n)$, then what does it imply? Explain. **[3]**
- c) What do Ω and Θ notations mean? When do we use O notation? **[6]**

OR

- Q4) a)** What is polynomial time reducibility? What is its importance in computational complexity theory? **[6]**
- b) What do understand by NP complete and NP hard problems? Give examples. **[6]**
- c) Is $6n^3 = \Theta(n^2)$? Justify your answer. **[3]**



Total No. of Questions : 4]

SEAT No. :

P8917

[Total No. of Pages : 2

Oct-22/BE/Insem-46

B.E. (Computer Engineering)

CYBER SECURITY AND DIGITAL FORENSICS

(2019 Pattern) (Semester - VII) (Elective - III) (410244C)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates :

- 1) Answer Q1 or Q2, Q3 or Q4.
- 2) Neat diagrams must be drawn whenever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

Q1) a) Define cybercrime. Explain types of cybercrime. [5]

b) Explain the process of security risk analysis. [5]

c) Give the reasons behind the need for information security. [5]

OR

Q2) a) Explain different threats to Information System. [5]

b) What do you mean by Cyber extortion, Drug trafficking? [5]

c) Describe in detail about cyber crime against an individual and organization. [5]

Q3) a) Explain in detail OS security. [5]

b) What are different types of viruses and worms? How do they propagate? [5]

c) What do you mean by Intellectual Property? [5]

OR

P.T.O.

- Q4)** a) Explain internet hacking and different approaches of hacking. [5]
- b) What are the different ways to gain access to your Computer System? [5]
- c) Explain the significance of Firewall and VPN security technologies. [5]

Total No. of Questions : 4]

SEAT No. :

P5194

[Total No. of Pages : 1

[6188]-146

B.E. (Computer Engineering) (Insem)
CYBER SECURITY & DIGITAL FORENSICS
(2019 Pattern) (Semester - VII) (Elective - III) (410244(C))

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Assume suitable data if necessary.

Q1) a) What are different threats to information system? Explain any two in details. [7]

b) List different types of cybercrime. Explain any two in detail. [8]

OR

Q2) a) Elaborate the process of risk analysis in cyber security. [7]

b) Write short note on : [8]

i) Cyber extortion

ii) Drug trafficking

Q3) a) What is mean by internet Hacking & Cracking? Explain types of cracking? [7]

b) What are different computer intrusion? Differentiate between virus & Worms. [8]

OR

Q4) a) Explain backups, archival storage & disposal of data in data security. [7]

b) Describe firewall security technology with neat diagram. [8]



Total No. of Questions : 4]

SEAT No. :

PC188

[Total No. of Pages : 1

[6361]-46

B.E. (Computer Engineering) (Insem)
CYBER SECURITY AND DIGITAL FORENSICS
(2019 Pattern) (Semester - VII) (Elective - III) (410244 (C))

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Attempt Q.No.1 or Q.No.2, Q.No.3 or Q.No.4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

- Q1)** a) What is cybercrime? How to prevent it? [5]
b) What is the need for Information Security? [5]
c) What are the types of Cyber Criminals? [5]

OR

- Q2)** a) What are the characteristics of Cyber Crime? [5]
b) What is cyber stalking? How it is conducted? [5]
c) Define Cyber Terrorism and state its objectives. [5]

- Q3)** a) What is Unauthorized Access? What are the common causes of Unauthorized Access? [5]
b) How to prevent unauthorized computer access? [5]
c) What is application security? State its different types. [5]

OR

- Q4)** a) How Email hacking takes place? [5]
b) What is Hardware Protection? What are different types of Hardware Protection? [5]
c) What are the types of Program Threats and system threats? [5]

